

COURSEWORK 2

COMPLETE MASTER GUIDE

Value-Sentiment Investment Strategy

Research | Code | Report Writing

Sector-Relative Scoring | Quality-Weighted Sentiment

Portfolio Construction | Backtesting | Performance

Target: 12,000–14,000 words | Deadline: 1 May 2026, 17:00 GMT

VERSION 3 — FINAL

Covers ALL task requirements: report, code, testing, config, reproducibility

TASK REQUIREMENTS CROSS-CHECK

Every requirement from the assignment brief mapped to this guide:

Task Requirement	Guide Section	Status
Investment Strategy Report (max 20k words)	Part B: full 11-section breakdown	Covered
Strategy Implementation (production-grade code)	Part D: code standards, architecture, config	Covered
Performance Summary (returns, vol, Sharpe, turnover)	Part A §A7 + Part B §8	Covered
Team Contributions & Reflections	Part B §11	Covered
Appendices (visualizations, flowcharts, tech docs)	Part C: checklists + Part D §D7	Covered
Clear and modular code structure	Part D §D1 (architecture)	Covered
Functions and classes	Part D §D2 (class design)	Covered
Meaningful variable/function names	Part D §D3 (naming conventions)	Covered
In-line comments	Part D §D3	Covered
Config separated from logic	Part D §D4 (backtest_config.yaml)	Covered
Version-controlled	Part D §D5 (Git workflow)	Covered
Reproducible from clean environment	Part D §D6 (setup guide)	Covered
Consistent with CW1 technical design	Part D §D7 (CW1 integration)	Covered
Selection rules (deciles, thresholds)	Part A §A4 + Part B §6	Covered
Weighting schemes (EW, factor, risk-parity)	Part A §A5	Covered
Reference benchmark	Part A §A7.3	Covered
Absolute and relative returns	Part A §A7.1	Covered
Risk-adjusted metrics	Part A §A7.1 (Sharpe, Sortino, etc)	Covered
Turnover measurement	Part A §A7.4	Covered
At least one robustness test	Part A §A8 (6 tests)	Covered
Strengths and weaknesses	Part B §10	Covered
Proposed improvements	Part B §10	Covered

REPORT STRUCTURE & WORD COUNTS

#	Section	Words	%	Assessment Criterion
1	Executive Summary	400–500	4%	Documentation (10%)
2	Introduction & Investment Rationale	1,500–1,800	13%	Investment Concept (25%)
3	Literature Review	1,200–1,500	11%	Investment Concept (25%)
4	Data & Universe (CW1 Integration)	800–1,000	7%	Teamwork (5%)
5	Signal Construction (UPGRADED)	1,500–1,800	13%	Methodology (30%)
6	Portfolio Construction Methodology	1,500–1,800	13%	Methodology (30%)
7	Backtesting Framework	1,000–1,200	9%	Methodology (30%)
8	Empirical Results & Performance	2,000–2,500	18%	Empirical Analysis (25%)
9	Robustness & Sensitivity Testing	1,500–1,800	13%	Empirical Analysis (25%)
10	Limitations, Future Work & Conclusions	800–1,000	7%	Documentation (10%)
11	Team Contributions & Reflections	400–500	4%	Teamwork (5%)

TOTAL: ~12,000–14,000 words (excluding references, appendices, tables, captions)

PART A: RESEARCH & METHODOLOGY

A1. Academic Foundations

Value Premium: Fama & French (1993, 2015) — persistent across markets. De Bondt & Thaler (1985) — behavioral overreaction. Lakonishok et al. (1994) — extrapolation. Zhang (2005) — distress risk. Value resurgence 2021–2022 supports sentiment timing.

Sentiment Effect: Baker & Wurgler (2006) — sentiment predicts cross-section. Stambaugh et al. (2012) — 11 anomalies stronger after high sentiment. Habibah et al. (2021) — sentiment as filter, not standalone. Supports 0.6/0.4 weighting.

VADER Accuracy: Kim et al. (2024): 44% on FOMC, ~56% on headlines. FinBERT: 88–93%. VADER is practical baseline; propose FinBERT as future improvement.

A2. Sector-Relative Value Scoring

UPGRADE: Replaces CW1 cross-sectional percentile ranking

Problem: Cross-sectional ranking makes financials/utilities always look 'cheap' vs tech. Ehsani, Harvey & Li (2023): standard HML 'consistently overweights Finance and Utilities'. This is an unintended sector bet.

Solution — MSCI 4-Stage Pipeline:

- **Stage 1:** Flip ratios (E/P, B/P, EBITDA/EV, Div Yield). Winsorize 2.5/97.5 pctl. Exclude EV/EBITDA for financials.
- **Stage 2:** Cross-sectional z-scores: $z = (x - \text{mean}) / \text{std}$ across all 605 stocks
- **Stage 3:** Composite $z = \text{average of metric z-scores}$. Then **within-sector re-standardization**:

$$Z_{\text{sector-rel}} = (Z_{\text{composite}} - \text{mean}_{\text{sector}}) / \text{std}_{\text{sector}}$$

- **Stage 4:** Cap at ± 3 . Use 11 GICS Sectors (~55 stocks each). Bayesian shrinkage for sectors <15 stocks.

Evidence: Ehsani et al. (2023): within-sector Sharpe 0.154 vs 0.032 across-sector. Asness, Porter & Stevens (2000): within-industry characteristics more precise.

```
# SECTOR-RELATIVE VALUE — key code
df['ep']=1.0/df['pe_ratio']; df['bp']=1.0/df['pb_ratio']
df['ebitda_ev']=1.0/df['ev_ebitda']
df.loc[df['gics_sector']=='Financials', 'ebitda_ev']=np.nan
# Winsorize, z-score, composite, then:
df['value_score'] = df.groupby('gics_sector')['comp_z']\
    .transform(lambda g: (g-g.mean())/g.std()).clip(-3,3)
```

A3. Quality-Weighted Sentiment Scoring

UPGRADE: Replaces CW1 volume-weighted scoring

Problem: volume_factor = min(count/20, 1) rewards noise. 50 wire copies $\neq$ 5x better signal. Tetlock (2011): stale reprints degrade signal.

Solution — 4-Component Quality Weight:

$$W_i = W_{\text{source}} \times W_{\text{relevance}} \times W_{\text{recency}} \times W_{\text{length}}$$

Component	Method	Range
Source credibility	Tier lookup: Reuters/Bloomberg=1.0, CNBC=0.7, SeekingAlpha=0.4, unknown=0.3	0.3–1.0
Relevance	Company in headline +0.5, body mentions +0.3, length $\geq$ 500w +0.2	0–1.0
Recency	Exponential decay: $e^{-(\ln 2/7 \times \text{days_old})}$, 7-day half-life	0–1.0
Substantiveness	min(word_count/500, 1.0)	0–1.0

Aggregation: $S = \text{Sum}(w_i \times \text{VADER}_i) / \text{Sum}(w_i)$. Then **consistency multiplier**: $c = \max(0, 1-2 \times \text{std})$. Then **Bayesian shrinkage**: $S_{\text{final}} = (n \times S_{\text{adj}}) / (n + 5)$. Confidence = $n/(n+5)$.

```
# QUALITY-WEIGHTED SENTIMENT – key code
arts['w'] = arts['w_src']*arts['w_rel']*arts['w_rec']*arts['w_len']
co = arts.groupby('ticker').apply(lambda g:
pd.Series({'sent': np.average(g['vader_compound'],weights=g['w']),
'std': np.sqrt(np.average((g['vader_compound']
-np.average(g['vader_compound'],weights=g['w']))**2,
weights=g['w']))), 'n':len(g)))
co['consistency']=(1-2*co['std']).clip(0)
co['final_sent']=(co['n']*co['sent']*co['consistency'])/(co['n']+5)
```

A4. Updated Composite Score & Screening

Composite = $0.6 \times \text{Value} + 0.4 \times \text{Sentiment}$ (same formula, upgraded inputs)

Scale alignment: Value is z-score (–3 to +3) → convert to percentile rank [0,100]. Sentiment is shrunk average (–1 to +1) → $(\text{final_sent}+1)/2 \times 100$.

Updated screening filters:

- Value score > 0 (above sector median)
- Sentiment confidence > 0.3 (~2+ quality-weighted articles)
- D/E < 2.0 (unchanged risk filter)
- Top 20% of eligible by Composite → invest_decision = TRUE

A5. Portfolio Construction

4 Portfolio variants (required by task: 'selection rules, weighting schemes, benchmark'):

Portfolio	Selection	Weighting	Purpose
A: Value-Only	Top 20% sector-rel value, D/E<2	Equal-weight	Isolate value
B: Sent-Only	Top 20% quality-weighted sent	Equal-weight	Isolate sentiment
C: Combined	invest_decision=TRUE	Equal-weight (primary)	Test hypothesis
D: Benchmark	S&P; 500 / MSCI World	Market-cap	Reference

3 Weighting schemes (task requires 'equal-weighted, factor-weighted, risk-parity, etc'):

- **Equal-weight:** $w=1/N$. DeMiguel et al. (2009): beats 14 optimization models. Primary.
- **Score-weighted:** $w=\text{CompositeScore}/\text{Sum}$. Tests signal strength. Secondary.
- **Inverse-volatility:** $w=(1/\text{sigma})/\text{Sum}(1/\text{sigma})$. 60-day trailing vol. Robustness check.

Constraints: Max 5% per stock, max 25% per sector, min 20 holdings, 30–50 target. Buffer rule: sell below 40th pctl, buy above 60th. Quarterly rebalancing (Jan 31, Apr 30, Jul 31, Oct 31).

A6. Backtesting Framework

Task requires: 'Implement the Backtesting Environment — use the existing data pipeline to simulate portfolio performance across time.'

Point-in-time discipline: report_date <= rebalance_date. 90-day lag buffer. T+1 close execution.

Survivorship bias: 73 delisted tickers (10.8%) in historical universe. Elton et al. (1996): 0.9–2.1% annual bias.

Transaction costs: 25 bps baseline, 50 bps stress test.

Return calculation with weight drift:

```
def compute_period_returns(weights, daily_returns):
    w = weights.copy(); port_rets = []
    for date in daily_returns.index:
        r = daily_returns.loc[date, w.index].fillna(0)
        port_rets.append((w * r).sum())
        w = w * (1 + r); w = w / w.sum()
    return pd.Series(port_rets, index=daily_returns.index)
```

A7. Performance Evaluation

Task requires: 'returns, volatility, Sharpe ratio, turnover, diversification metrics'

A7.1 Return & Risk Metrics

Metric	Formula	Good Value
Ann. Return	$(1+R_{\text{total}})^{(252/\text{days})}-1$	—
Ann. Volatility	$\text{sigma_daily} \times \text{sqrt}(252)$	—
Sharpe	$(R_p - R_f) / \text{sigma}_p$	>1.0
Sortino	$(R_p - R_f) / \text{sigma_downside}$	>2.0
Calmar	$\text{Ann.Return} / \text{MaxDD} $	>1.0
Information Ratio	$(R_p - R_b) / \text{TE}$	>0.5
Max Drawdown	$\min((\text{cum} - \text{running_max}) / \text{running_max})$	—

A7.2 Fama-French Factor Regression

$R_p - R_f = \alpha + b_{\text{MKT}}(R_m - R_f) + b_{\text{SMB}}(\text{SMB}) + b_{\text{HML}}(\text{HML}) + b_{\text{RMW}}(\text{RMW}) + b_{\text{CMA}}(\text{CMA}) + e$. Use Newey-West SE (6 lags). Data from Kenneth French's library.

A7.3 Benchmark Selection

Task: 'Clearly identify a reference benchmark used to evaluate KPI and Risk'

- **Primary:** S&P 500 (^GSPC via yfinance) — most recognized, easy data access
- **Secondary:** MSCI World Value (via IWVL.L ETF) — captures value tilt
- **Equal-weight universe:** all 605 stocks equally weighted — shows selection vs market

A7.4 Turnover Calculation

Task explicitly requires turnover measurement.

$\text{Turnover} = \text{Sum}(|w_{\text{new},i} - w_{\text{old},i}|) / 2$ (one-way, per rebalance)

```
def calc_turnover(new_w, old_w):
    all_tickers = new_w.index.union(old_w.index)
    nw = new_w.reindex(all_tickers, fill_value=0)
    ow = old_w.reindex(all_tickers, fill_value=0)
    return (nw - ow).abs().sum() / 2
```

Report: average quarterly turnover, annual turnover, cumulative cost impact.

A7.5 Diversification Metrics

Task explicitly requires diversification metrics.

- **HHI:** $\text{Sum}(w_i^2)$. EW 40-stock = 0.025. Lower = more diversified.

- **Effective N:** $1/HHI$. Report at each rebalance.
- **Sector concentration:** max sector weight, sector count
- **VaR (95%):** `np.percentile(returns, 5)`
- **CVaR (95%):** mean of returns below VaR (coherent risk measure)

A8. Robustness & Sensitivity Testing

Task requires: 'at least one robustness or sensitivity test'. We do 6:

Test	Method	What It Shows
1. Weight sensitivity	Vary 0.6/0.4 from 0/100 to 100/0 in 5% steps	Is 60/40 near-optimal?
2. Threshold sensitivity	Top 10/15/20/25/30%, D/E 1.5/2.0/2.5/3.0	How concentrated?
3. Sub-period analysis	Year-by-year + 2021-23 vs 2023-25	Consistent across regimes?
4. Bootstrap CIs	Stationary bootstrap, 2500 reps, 95% CI	Is Sharpe > 0?
5. Random portfolios	10,000 random of same size, Sharpe histogram	Skill vs luck?
6. Sector attribution	Leave-one-sector-out, re-run backtest	Driven by one sector?

PART B: SECTION-BY-SECTION REPORT WRITING GUIDE

Section 1: Executive Summary (400–500 words)

Write LAST. Strategy summary, hypothesis, headline numbers (Sharpe, return, MaxDD vs benchmark), key finding. No citations.

Section 2: Introduction & Investment Rationale (1,500–1,800 words)

- **2.1 Value Trap Problem** (400w) — cite Graham & Dodd, Fama & French, Lakonishok et al.
- **2.2 Sentiment as Behavioural Filter** (400w) — cite Baker & Wurgler, Tetlock (2007, 2008)
- **2.3 Combined Strategy + Sector-Relative Innovation** (400w) — cite Ehsani et al., Stambaugh et al.
- **2.4 Hypotheses** (200w): H1 (higher Sharpe), H2 (lower drawdown), H3 (60/40 near-optimal)

Section 3: Literature Review (1,200–1,500 words)

Narrative argument: (3.1) Value Premium, (3.2) Sentiment, (3.3) NLP/VADER limits, (3.4) Combined strategies, **(3.5) Sector neutralization** (NEW — Ehsani et al., Asness et al. 2000), **(3.6) News quality** (NEW — Tetlock 2011). End every paragraph connecting to YOUR strategy.

Section 4: Data & Universe (800–1,000 words)

Bridge CW1→CW2. Architecture recap (PostgreSQL/MongoDB/MinIO/Kafka). 605 active, 73 delisted. 82–89% coverage. Benchmark data. Include data coverage table.

Section 5: Signal Construction (1,500–1,800 words)

UPGRADE: Substantially richer than CW1 — this is where you show CW2 innovation

- **5.1 Sector-Relative Value** (600w): MSCI 4-stage, formulas, GICS justification, shrinkage
- **5.2 Quality-Weighted Sentiment** (600w): 4 components, consistency, Bayesian shrinkage
- **5.3 Composite Score** (300w): $0.6V + 0.4S$, scale alignment, updated filters
- **Include:** formula summary table, OLD vs NEW comparison table, validation results

Section 6: Portfolio Construction (1,500–1,800 words)

4 portfolio variants, 3 weighting schemes, constraints (5%/25%), rebalancing, transaction costs. Cite DeMiguel et al. (2009) for equal-weight justification.

Section 7: Backtesting Framework (1,000–1,200 words)

Point-in-time, survivorship, return calculation with drift, pitfalls table. Cite Bailey et al. (2015), Lopez de Prado (2018).

Section 8: Empirical Results (2,000–2,500 words)

THE critical section. Summary performance table (all 4 portfolios × all metrics), cumulative return chart, drawdown analysis, FF regression, sector analysis, weighting comparison, monthly heatmap. **Always show t-statistics for alpha.**

Section 9: Robustness (1,500–1,800 words)

All 6 tests from A8. Weight sensitivity heatmap, threshold table, sub-period, bootstrap CIs, random portfolios histogram, sector attribution.

Section 10: Limitations & Conclusions (800–1,000 words)

Task: 'identify strengths and weaknesses, propose potential improvements'

- **Limitations:** survivorship bias, VADER 56%, GDELT coverage, short backtest, long-only
- **Improvements:** FinBERT, historical universe, dynamic timing, momentum factor, 10yr backtest
- **Conclusion:** restate H1/H2/H3 results, headline numbers, contribution

Section 11: Team Contributions (400–500 words)

Task: 'collaborative workflow, team coordination, lessons learned'

- Role assignments: Product Owners, Specialists, Developers
- CW1→CW2 integration: how pipeline feeds backtester
- Challenges and lessons learned

PART C: CHARTS, TABLES & APPENDICES CHECKLISTS

C1. Essential Charts (12)

#	Chart	Section
1	Cumulative return (4 portfolios + benchmark, log scale)	8
2	Drawdown underwater chart	8
3	Monthly returns heatmap (12×N years)	8
4	Rolling 12-month Sharpe ratio	8
5	Weight sensitivity heatmap (value/sentiment → Sharpe)	9
6	Fama-French factor loadings with CIs	8
7	Sector allocation (portfolio vs benchmark)	8
8	Random portfolio Sharpe histogram (10,000)	9
9	Screening threshold sensitivity	9
10	Turnover per rebalance date	8
11	OLD vs NEW value score sector concentration	5
12	Pipeline flowchart (CW1→CW2)	4/App

C2. Essential Tables (11)

#	Table	Section
1	Summary performance (all portfolios × all metrics)	8
2	Fama-French 5-factor regression	8
3	Year-by-year returns and Sharpe	9
4	Weight sensitivity results	9
5	Threshold sensitivity	9
6	Weighting scheme comparison	8
7	Top 3 drawdown events	8
8	Bootstrap 95% CIs	9
9	OLD vs NEW value scoring comparison	5
10	OLD vs NEW sentiment scoring comparison	5
11	Backtesting pitfalls addressed	7

C3. Appendices (Task: 'visualizations, flowcharts, technical documentation')

- **Appendix A:** Full code architecture diagram + module descriptions
- **Appendix B:** Complete backtest results (all months, all portfolios)
- **Appendix C:** Additional robustness tables (all parameter combinations)
- **Appendix D:** QuantStats tearsheet (HTML→PDF)
- **Appendix E:** Mermaid/flowchart diagrams
- **Appendix F:** Data quality summary (from CW1, updated)
- **Appendix G:** Code quality metrics (test coverage, linting)
- **Appendix H:** Configuration file (backtest_config.yaml — full contents)

- **Appendix I:** Installation & reproduction guide

PART D: TECHNICAL IMPLEMENTATION GUIDE

This part addresses the task requirement: 'All code must meet production-grade standards'

D1. Code Architecture (Task: 'clear and modular structure')

```
team_09/
coursework_one/ # CW1 (unchanged)
modules/
config/conf.yaml
Main.py
coursework_two/ # CW2 (new)
config/
backtest_config.yaml # ALL parameters here
modules/
data/
data_loader.py # Load from CW1's PostgreSQL/MongoDB
universe.py # Point-in-time universe construction
benchmark.py # Benchmark data retrieval
signals/
value_signal.py # Sector-relative z-scores
sentiment_signal.py # Quality-weighted aggregation
signal_combiner.py # 0.6*value + 0.4*sentiment
portfolio/
portfolio_constructor.py # Screen->weight->constrain
constraints.py # Sector caps, position limits
weighting.py # EW, score-weight, inv-vol
backtest/
backtester.py # Main loop
transaction_costs.py
rebalance_schedule.py
analytics/
performance.py # Sharpe, Sortino, drawdown
risk.py # VaR, CVaR, FF regression
turnover.py # Turnover calculation
diversification.py # HHI, effective N, sector conc
robustness/
sensitivity.py # Weight/threshold grids
bootstrap.py # Stationary bootstrap CIs
random_portfolios.py
visualization/
charts.py # All 12 charts
tearsheet.py # QuantStats integration
tests/
test_value_signal.py
test_sentiment_signal.py
test_portfolio.py
test_backtester.py
test_performance.py
Main_CW2.py # Single entry point
README.md
```

D2. Class Design (Task: 'appropriate use of functions and classes')

```
class Backtester:
    """Orchestrates quarterly backtest loop."""
    def __init__(self, universe, signal_gen, port_con, cost_model):
        self.universe = universe
        self.signal_gen = signal_gen
        self.port_con = port_con
        self.cost_model = cost_model

    def run(self, start, end, prices):
        rebal_dates = self._get_rebalance_dates(start, end)
        all_returns, current_w = [], pd.Series(dtype=float)
        for i, rd in enumerate(rebal_dates):
            univ = self.universe.get_universe(rd)
            sigs = self.signal_gen.compute(rd, univ)
            new_w = self.port_con.construct(sigs)
            tc = self.cost_model.calculate(current_w, new_w)
            next_rd = rebal_dates[i+1] if i+1<len(rebal_dates) else end
```

```
rets = self._period_returns(new_w, prices, rd, next_rd)
rets.iloc[0] -= tc
all_returns.append(rets)
current_w = self._drift_weights(new_w, rets)
return pd.concat(all_returns)

class PortfolioConstructor:
    """Screen, weight, apply constraints."""
    def __init__(self, config):
        self.max_position = config['max_position']
        self.max_sector = config['max_sector_weight']
        self.scheme = config['weighting_scheme']

    def construct(self, signals):
        eligible = self._screen(signals)
        weights = self._weight(eligible)
        return self._apply_constraints(weights)
```

D3. Naming Conventions & Comments (Task: 'meaningful names, in-line comments')

Rule	Good Example	Bad Example
Functions: verb_noun	compute_value_scores()	scores()
Variables: descriptive	sector_relative_zscore	srz
Constants: UPPER_SNAKE	MAX_POSITION_WEIGHT = 0.05	maxw = 0.05
Classes: PascalCase	PortfolioConstructor	portcon
Files: snake_case	value_signal.py	ValueSignal.py
Boolean: is/has prefix	is_eligible, has_sentiment	eligible, sent

Docstring standard (same as CW1 — Sphinx :param:/:returns:/:raises:):

```
def compute_sector_relative_value(df, date):
    """Compute within-sector z-scored value for all companies.

    Implements MSCI Enhanced Value 4-stage pipeline:
    flip ratios, cross-sectional z, composite, sector restand.

    :param df: DataFrame with pe_ratio, pb_ratio, ev_ebitda,
    dividend_yield, gics_sector columns.
    :type df: pd.DataFrame
    :param date: Reference date (YYYY-MM-DD).
    :type date: str
    :returns: DataFrame with value_score column (-3 to +3).
    :rtype: pd.DataFrame
    :raises ValueError: If no valid ratios for any company.
    """
```

D4. Configuration (Task: 'separation of configuration from logic')

backtest_config.yaml — ALL tuneable parameters in one file, never hardcoded:

```
# ■■■ backtest_config.yaml ■■■
backtest:
  start_date: '2021-01-31'
  end_date: '2025-12-31'
  rebalance_frequency: quarterly # quarterly|monthly|weekly
  rebalance_months: [1, 4, 7, 10] # End of Jan/Apr/Jul/Oct
  reporting_lag_days: 90
  execution_delay: 1 # T+1 close

scoring:
  value_weight: 0.6
  sentiment_weight: 0.4
  max_debt_equity: 2.0
  min_sentiment_confidence: 0.3
  selection_percentile: 0.20 # Top 20%
  winsorize_lower: 0.025
  winsorize_upper: 0.975
  zscore_cap: 3.0
  shrinkage_k_sector: 20 # Bayesian shrinkage for small sectors
  shrinkage_k_sentiment: 5

sentiment:
  half_life_days: 7
  source_tiers: # Loaded from separate YAML or dict
  tier1: ['reuters.com', 'bloomberg.com', 'ft.com', 'wsj.com']
  tier2: ['cnbc.com', 'barrons.com', 'bbc.com', 'marketwatch.com']
  tier3: ['seekingalpha.com', 'benzinga.com', 'fool.com']
  tier_weights: {tier1: 1.0, tier2: 0.7, tier3: 0.4, default: 0.3}

portfolio:
  weighting_scheme: equal_weight # equal_weight|score|inv_vol
  max_position_weight: 0.05
  max_sector_weight: 0.25
  min_holdings: 20
  target_holdings: 40
  buffer_buy_pctl: 0.60
  buffer_sell_pctl: 0.40
```

```
costs:
transaction_cost_bps: 25 # one-way
stress_test_bps: 50

benchmark:
primary: '^GSPC' # S&P; 500
secondary: 'IWL.L' # MSCI World Value ETF

data:
postgres_table_prices: 'systematic_equity.daily_prices'
postgres_table_values: 'systematic_equity.value_metrics'
postgres_table_sentiment: 'systematic_equity.sentiment_scores'
postgres_table_composite: 'systematic_equity.composite_rankings'
postgres_table_company: 'systematic_equity.company_static'
lookback_years: 5
```

Loading config:

```
import yaml
with open('config/backtest_config.yaml') as f:
    config = yaml.safe_load(f)
# Access: config['scoring']['value_weight'] → 0.6
```


D5. Version Control (Task: 'version-controlled')

Git workflow consistent with CW1:

- **main** branch: stable, submission-ready code
- **develop** branch: integration branch
- **feature/** branches: feature/backtester, feature/value-signal, feature/charts
- Pull requests with code review before merging
- Commit convention: **feat:**, **fix:**, **docs:**, **test:** prefixes

```
# Example commits:
feat: implement sector-relative value scoring
feat: add quality-weighted sentiment aggregation
feat: backtester with quarterly rebalancing
test: add known-answer tests for portfolio returns
docs: add Sphinx docstrings to all public functions
fix: handle NaN in EV/EBITDA for financials
```

D6. Reproducibility (Task: 'reproducible from clean environment')

Step-by-step setup guide (include in README.md):

```
# 1. Prerequisites: Docker Desktop, Python 3.10+, Poetry 1.7+

# 2. Clone and navigate
git clone https://github.com/.../ift_coursework_2025.git
cd ift_coursework_2025/team_09

# 3. Start CW1 infrastructure (required for data)
cd coursework_one && docker compose up -d
# Wait for postgres-seed, mongo-seed to finish (exit 0)

# 4. Verify database
docker exec postgres-db psql -U postgres -d ift -c \
'SELECT COUNT(*) FROM systematic_equity.company_static;'
# Expected: 678

# 5. Run CW1 pipeline to populate data
cd coursework_one && poetry install
poetry run python Main.py --env_type dev --frequency quarterly

# 6. Install CW2 dependencies
cd ../coursework_two && poetry install

# 7. Run CW2 backtest
poetry run python Main_CW2.py --config config/backtest_config.yaml

# 8. Run tests
poetry run pytest tests/ -v --cov=modules

# 9. Generate report charts
poetry run python -m modules.visualization.charts
```

pyproject.toml dependencies (CW2-specific additions):

```
[tool.poetry.dependencies]
python = '^3.10'
pandas = '^2.0'
numpy = '^1.24'
scipy = '^1.10'
statsmodels = '^0.14'
matplotlib = '^3.7'
seaborn = '^0.12'
pyyaml = '^6.0'
psycpg2-binary = '^2.9'
yfinance = '^0.2'
quantstats = '^0.0.62'
arch = '^6.0' # for bootstrap
```

D7. CW1 Integration (Task: 'consistent with CW1 technical design')

How CW2 connects to CW1:

CW1 Output	CW2 Usage
company_static table (678 companies)	Universe definition + GICS sectors
daily_prices table	Daily returns for backtest
value_metrics table	Quarterly P/E, P/B, EV/EBITDA, Div Yield, D/E
sentiment_scores table	Article-level VADER scores
composite_rankings table	CW1 baseline for comparison
MongoDB raw_news_articles	Source domain, headline, word count for quality weights
conf.yaml database connections	Shared connection strings
pipeline_metadata table	Last run dates for point-in-time logic

```
# data_loader.py – connect to CW1's PostgreSQL
import psycopg2, yaml
with open('../coursework_one/config/conf.yaml') as f:
    cw1_config = yaml.safe_load(f)
conn = psycopg2.connect(
    host=cw1_config['dev']['postgres']['host'],
    port=cw1_config['dev']['postgres']['port'],
    dbname=cw1_config['dev']['postgres']['database'],
    user=cw1_config['dev']['postgres']['user'],
    password=cw1_config['dev']['postgres']['password'])
```

D8. Testing Strategy (extending CW1's 582 tests)

Test Type	What	Example
Known-answer	Manual calculation, verify exact match	2-stock 2-day portfolio → return = X
Invariants	Weights sum to 1, ≥0, ≤5%	assert abs(weights.sum()-1) < 1e-10
No-lookahead	Signals use only pre-rebalance data	assert all(signal_dates < rebalance_date)
Edge cases	Empty universe, single stock, all NaN	Empty → raise, single → w=1.0
Regression	Saved reference output unchanged	Compare to saved CSV after refactor
Integration	Full pipeline from DB to results	Run mini-backtest on 3 stocks, 2 quarters

```
# Example test
def test_equal_weight_sums_to_one():
    signals = pd.DataFrame({'composite': [80, 60, 70]},
                           index=['AAPL', 'MSFT', 'GOOGL'])
    w = portfolio_constructor.construct(signals)
    assert abs(w.sum() - 1.0) < 1e-10
    assert (w >= 0).all()
    assert (w <= 0.05 + 1e-10).all() # max 5%

def test_no_lookahead():
    rebal = '2023-04-30'
    scores = signal_gen.compute('2023-04-30', universe)
    # Check that no data after rebal was used
    for col in scores.columns:
        assert scores[col+'_date'].max() <= rebal
```

Coverage target: 85%+ across all CW2 modules (CW1 achieved 91%).

PART E: REFERENCES & IMPLEMENTATION ROADMAP

E1. Complete Reference List (42 Sources)

Value & Factor Models

1. Asness et al. (2013) 'Value and momentum everywhere', *JoF*, 68(3).
2. Asness, Porter & Stevens (2000) 'Industry-relative characteristics', SSRN 213872.
3. Blitz & Hanauer (2020) 'Resurrecting the value premium', *JPM*, 47(2).
4. De Bondt & Thaler (1985) 'Does the stock market overreact?', *JoF*, 40(3).
5. Ehsani, Harvey & Li (2023) 'Is sector neutrality a mistake?', *FAJ*, 79(4).
6. Fama & French (1993) 'Common risk factors', *JFE*, 33(1).
7. Fama & French (2015) 'Five-factor model', *JFE*, 116(1).
8. Graham & Dodd (1934) *Security analysis*.
9. Lakonishok, Shleifer & Vishny (1994) 'Contrarian investment', *JoF*, 49(5).
10. Zhang (2005) 'The value premium', *JoF*, 60(1).

Sentiment & NLP

11. Baker & Wurgler (2006) 'Investor sentiment and cross-section', *JoF*, 61(4).
12. Hutto & Gilbert (2014) 'VADER', *ICWSM-14*.
13. Kim et al. (2024) 'Evaluating language models for finance', *ACM ICAIF*.
14. Stambaugh, Yu & Yuan (2012) 'Sentiment and anomalies', *JFE*, 104(2).
15. Tetlock (2007) 'Giving content to investor sentiment', *JoF*, 62(3).
16. Tetlock et al. (2008) 'More than words', *JoF*, 63(3).
17. Tetlock (2011) 'All the news that's fit to reprint', *RFS*, 24(5).
18. Habibah et al. (2021) 'Sentiment and factor premia', *JES*, 48(5).

Portfolio, Risk & Backtesting

19. Bailey et al. (2015) 'Probability of backtest overfitting', *JCF*, 20(4).
20. DeMiguel et al. (2009) 'Optimal vs naive diversification', *RFS*, 22(5).
21. Elton et al. (1996) 'Survivorship bias', *RFS*, 9(4).
22. Lo (2002) 'Statistics of Sharpe ratios', *FAJ*, 58(4).
23. Lopez de Prado (2018) *Advances in financial machine learning*.
24. Maillard et al. (2010) 'Risk contribution portfolios', *JPM*, 36(4).
25. Newey & West (1987) 'HAC covariance', *Econometrica*, 55(3).
26. Politis & Romano (1994) 'Stationary bootstrap', *JASA*, 89(428).
- 27–42. [Additional: MSCI methodology, S&P methodology, RavenPack, Araci (2019), etc.]

E2. Implementation Roadmap (4 Weeks)

Week	Focus	Deliverables
1 (Apr 1–7)	Signals + portfolio code	value_signal.py, sentiment_signal.py, portfolio_constructor.py, constraints.py, backtest_config.yaml, unit tests
2 (Apr 8–14)	Backtester + full run	backtester.py, transaction_costs.py, performance.py, turnover.py — run all 4 portfolios, get results
3 (Apr 15–21)	Analysis + robustness	risk.py, sensitivity.py, bootstrap.py, visualization/ — all 12 charts, 11 tables
4 (Apr 22–30)	Report + polish	Write report sections 1–11. Code docs, README, team reflections. Submit May 1 17:00 GMT

E3. Priority If Time Is Tight

- 1. value_signal.py (sector-rel) + backtester + equal-weight + summary table → core 30% marks

- **2.** sentiment_signal.py (quality) → shows CW2 innovation vs CW1
- **3.** Benchmark comparison + cumulative return chart → context
- **4.** Weight sensitivity heatmap → highest-impact robustness test
- **5.** FF regression → quant sophistication
- **6.** Everything else in decreasing assessment weight

E4. Writing Style

- Active voice. Precise numbers. Every claim needs evidence.
- Be honest about underperformance. Link to H1/H2/H3.
- Past tense for results. Present tense for theory.
- Caption every figure and table. Cross-reference sections.
- Cut filler words: 'very', 'really', 'quite', 'somewhat'.

v3 FINAL — This guide covers ALL task requirements: research (Part A), report writing (Part B), checklists (Part C), technical implementation including code standards, config, Git, testing, reproducibility, and CW1 integration (Part D), plus references and roadmap (Part E). Good luck!